BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6191

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. (7, 9) + (3, –5) = :

(A) (7, 9)
(B) (3, –5)
(C) (10, 4)
(D) (4, 10)

Q2. Which cannot be used as binary operation?

(A) Addition '+'
(B) Division '÷'
(C) Multiplication '×'
(D) Square root '√'

Q3. If adjoint of a matrix A = [–1 –2; 3 4], then matrix A is:

(A) [–1 –2; 4 3]
(B) [–4 3; –2 1]
(C) [4 2; 3 –1]
(D) [4 2; –3 –1]

Q4. Rank of the matrix [1; 1; 1; 0] is:

(A) 0
(B) 1
(C) 2
(D) 3

Q5. Which equation has roots 2 and –3 ?

(A) $x^2 + x + 6 = 0$ (B) $x^2 + x - 6 = 0$

(C) $x^2 - x - 6 = 0$ (D) $x^2 - x + 6 = 0$

Q6. If roots of the equation $x^2 + px + q = 0$ are additive inverse of each other, then which is true?

(A) $p = 0$ (B) $q = 1$

(C) $p = 1$ (D) $p = q$

Q7. $(x^2 + x - 1)/Q(x)$ will be an improper fraction, if:

(A) Degree of Q(x) = 2 (B) Degree of Q(x) > 2

(C) Degree of Q(x) = 3 (D) Degree of Q(x) ≠ 2

Q8. Sum of 5 A.Ms between 2 and 8 is:

(A) 10 (B) 40

(C) 25 (D) 50

Q9. If a and b are negative distinct real numbers, then with usual notations, which is correct?

(A) A > G (B) H < G

(C) A < G (D) A > G > H

Q10. Which cannot be the term of a G.P. ?

(A) –1 (B) 0

(C) 1 (D) 5

Q11. A coin is tossed 5 times, then total number of outcomes n(S) = :

(A) 10 (B) 25

(C) 20 (D) 32

Q12. 2nd term in the expansion of $(1 - x)^{-1}$ is:

(A) 1 (B) 2x

(C) x (D) –x

Q13. $\sec\theta \cdot \operatorname{cosec}\theta \cdot \sin\theta \cdot \cos\theta =$

(A) 1 (B) –1

(C) 0 (D) Cannot be determined

Q14. In a right angled triangle, the side opposite to right angle is called:

(A) Base (B) Hypotenuse

(C) Perpendicular (D) Altitude

Q15. If $\sin \alpha = 2/3$, $\cos \alpha = 3/4$, then value of $\sin 2\alpha$ = :

(A) 1/2 (B) 1/3

(C) 15/144 (D) 1

Q16. Period of $2 + \cos 3x$ is:

(A) π (B) $3\pi/2$

(C) 2π (D) $2\pi/3$

Q17. In any triangle ABC, $a = 4$, $b = 10$, $\gamma = 30°$, then area of triangle Δ = :

(A) 5 sq.units (B) 20 sq.units

(C) 10 sq.units (D) 40 sq.units

Q18. If a, b and c are the sides of a triangle ABC, then $(c^2 + b^2 - a^2)/2bc$ = :

(A) $\cos \alpha$ (B) $\cos \gamma$

(C) $\cos \beta$ (D) $\cos^2 \alpha$

Q19. If $\sin^{-1}a = 0$, then value of a is:

(A) $\pi/2$ (B) π

(C) 0 (D) 0 & π

Q20. The solution of the equation $2 \sin x + \sqrt{3} = 0$ in 4th quadrant is:

(A) $\pi/3$ (B) $5\pi/3$

(C) $-\pi/4$ (D) $-\pi/6$

Mathematics (Subjective) Group - I

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 6191

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) Does the set { 1 , –1 } possess closure property with respect to addition and subtraction?

(ii) Find the difference and product of the complex numbers (8 , 9) and (5 , –6)

(iii) Find the multiplicative inverse of ($\sqrt{2}$, $+\sqrt{5}$)

(iv) If U = { 1, 2, 3, 4, 5, ... , 20 } and A = { 1, 3, 5, ... , 19 } verify $A \cup A' = U$

(v) Write the inverse and contrapositive of the conditional $\sim q \rightarrow \sim p$

(vi) If a , b are elements of a group G then show that $(ab)^{-1} = b^{-1}a^{-1}$

(vii) Find x and y if $[x+3\ 1; -3\ 3y-4] = [2\ 1; -3\ 2]$

(viii) If A and B are square matrices of the same order, then explain why in general $(A + B)(A - B) \neq A^2 - B^2$

(ix) Without expansion verify that $|\alpha\ \beta+\gamma\ 1;\ \beta\ \gamma+\alpha\ 1;\ \gamma\ \alpha+\beta\ 1| = 0$

(x) Use the remainder theorem to find remainder when first polynomial is divided by second polynomial $x^2 + 3x + 7$, $x + 1$

(xi) Find the condition that one root of the equation $x^2 + px + q = 0$ is multiplicative inverse of the other.

(xii) Discuss the nature of roots of the equation $2x^2 + 5x - 1 = 0$

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Resolve $(3x - 11)/((x^2 + 1)(x + 3))$ into partial fractions without finding constants.

(ii) Change $(6x^3 + 5x^2 - 7)/(2x^2 - x - 1)$ into proper rational fraction.

(iii) Find the indicated term of the sequence 1, –3, 5, –7, 9, –11, ... , a_8

(iv) If $a_{n-3} = 2n - 5$ find the nth term of the sequence.

(v) Find the nth term of the geometric sequence if a_5/a[illegible in source scan] = 4/9 and a_2 = 4/9

(vi) Find A, G, H and verify that $A > G > H$ ($G > 0$) if $a = 2$ and $b = 8$.

(vii) Write $n(n - 1)(n - 2) \ldots (n - r + 1)$ in factorial form.

(viii) Find the value of n when ${}^nP_2 = 30$

(ix) A die is rolled. What is the probability that the dots on the top are greater than 4.

(x) Using binomial theorem expand $(a + 2b)^5$.

(xi) Expand (1 – x)^(1/2) up to 4 terms.

(xii) Using binomial theorem to find the values of √99 to three places of decimals.

Q4. (Marks 18) Attempt any NINE parts:

(i) Find ℓ, when $\theta = \pi$ radians , $r = 6$cm

(ii) Verify $\sin^2 \pi/6 + \sin^2 \pi/3 + \tan^2 \pi/4 = 2$

(iii) Prove that $\sec^2\theta - \operatorname{cosec}^2\theta = \tan^2\theta - \cot^2\theta$

(iv) Without using the table, find the value of sin(–300°)

(v) Prove that $\cos(\alpha + 45°) = 1/\sqrt{2}\ (\cos\alpha - \sin\alpha)$

(vi) Prove that $\cot\alpha - \tan\alpha = 2\cot 2\alpha$

(vii) Find the period of tan x/7

(viii) Find the greatest angle of triangle ABC if a = 16 , b = 20 , c = 33

Note: (Continued P/2)

(ix) Find area of triangle ABC, given sides are a = 18 , b = 24 , c = 30

(x) Prove that $r_1\ r_2\ r_3 = rs^2$

(xi) Find the value of $\tan(\cos^{-1}\sqrt{3}/2)$

(xii) Find the solution of the equation $\sin x = -\sqrt{3}/2$ which lies in $[0 , 2\pi]$

(xiii) Solve $\cot\theta = 1/\sqrt{3}$ where $\theta \in [0 , 2\pi]$

Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5(a). (Marks 05) Show that $|b+c\ a\ a^2;\ c+a\ b\ b^2;\ a+b\ c\ c^2| = (a + b + c)(a - b)(b - c)(c - a)$

Q5(b). (Marks 05) If α , β are the roots of equation $ax^2 + bx + c = 0$ then form the equation whose roots are $1/\alpha^3 , 1/\beta^3$

Q6(a). (Marks 05) Resolve into partial fractions: $9/((x + 2)^2(x - 1))$

Q6(b). (Marks 05) If the H.M and A.M between two numbers are 4 and 9/2 respectively, find the numbers.

Q7(a). (Marks 05) Prove that: ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^{n}C_r$

Q7(b). (Marks 05) Determine the middle term in the expansion of $(1/x - x^2/2)^{12}$

Q8(a). (Marks 05) If $\cot\theta = 5/2$ and the terminal arm of the angle is in the I-quadrant, find the values of $(3\sin\theta + 4\cos\theta)/(\cos\theta - \sin\theta)$

Q8(b). (Marks 05) Prove that $(2\sin\theta\sin 2\theta)/(\cos\theta + \cos 3\theta) = \tan 2\theta \tan\theta$

Q9(a). (Marks 05) Show that $1/r^2 + 1/r_1^2 + 1/r_2^2 + 1/r_3^2 = (a^2 + b^2 + c^2)/\Delta^2$

Q9(b). (Marks 05) Prove that $2\tan^{-1} 2/3 = \sin^{-1} 12/13$

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6196

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero

marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. For any two real numbers a and b, G^2 is equal to:

(A) AH (B) H/A

(C) A/H (D) √AH

Q2. If 1/5, 1/8 are two harmonic means between a and b, then value of b is:

(A) 1/3 (B) 1/10

(C) 1/11 (D) 1/13

Q3. If $a_n = n + (-1)^n$, then a_{10} = :

(A) 10 (B) 11

(C) 9 (D) –11

Q4. $P(x)/(x^2 + 1)$ is proper fraction, if degree of the polynomial P(x) is:

(A) Equal to 2 (B) Greater than 2

(C) Not equal to 2 (D) Less than 2

Q5. Degree of a constant polynomial is:

(A) 1 (B) 0

(C) 2 (D) Arbitrary

Q6. If one root of $x^2 + ax + 2 = 0$ is 2, then value of a is:

(A) –3 (B) 4

(C) 3 (D) –2

Q7. If A = [2 1; 6 3], then cofactor of 6 is:

(A) 1 (B) –6

(C) –1 (D) 3

Q8. The matrix [7] is:

(A) Row matrix (B) Square matrix

(C) Column matrix (D) All these

Q9. If ~p → q is a conditional, then its converse is:

(A) q → ~p (B) ~q → p

(C) p → ~q (D) ~q → ~p

Q10. If r is the radius and C is the circumference of a circle, then value of C/r = :

(A) π (B) π/2

(C) 2π (D) 1/(2π)

Q11. The solution of equation tan x = 1/√3 lies in quadrants:

(A) I & II
(B) I & III
(C) II & IV
(D) I & IV

Q12. If x = $\sin^{-1}$ √3/2, then value of x is:

(A) –π/2
(B) –π/3
(C) π/3
(D) π/6

Q13. With usual notations, abc/Δ = :

(A) 4R
(B) r
(C) R
(D) rs

Q14. In any triangle ABC, if two sides and their included angle is given, then area of triangle is:

(A) Δ = 1/2 bc sin α
(B) 1/2 ab sin γ
(C) Δ = 1/2 ac sin β
(D) All these

Q15. Period of 2 cosec x/4 is:

(A) π/2
(B) 4π
(C) 2π
(D) 8π

Q16. Value of sin 7π is equal to:

(A) 1
(B) 1/2
(C) –1
(D) 0

Q17. Angle 5π/9 lies in quadrant:

(A) I
(B) III
(C) II
(D) IV

Q18. If ℓ = 1.5cm, r = 2.5cm, then value of θ is:

(A) 3.75 rad
(B) 3/5 rad
(C) 0.60 rad
(D) 5/3 rad

Q19. The 2nd term in the expansion of (1 – 2x)^(1/3) is:

(A) –2/3 x
(B) 2/3 x
(C) 4/9 x^2
(D) 3/2 x

Q20. The number of permutations of the word PANAMA are:

(A) 10
(B) 60
(C) 20
(D) 120

Mathematics (Subjective) Group - II

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 6196

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) Simplify (–1)^(–21/2)

(ii) Show that $z\bar{z} = |z|^2$

(iii) Find multiplicative inverse of –3 – 5i

(iv) Find converse and inverse of $\sim p \rightarrow \sim q$

(v) Write $\{ x \mid x \in 0 \wedge 3 < x < 12 \}$ in descriptive and tabular form.

(vi) Show that subtraction is non-commutative on 'N'.

(vii) Find x and y if [x+3 1; –3 3y–4] = [2 1; –3 2]

(viii) Find inverse of [–2 3; –4 5]

(ix) Without expansion show that |6 7 8; 3 4 5; 2 3 4| = 0

(x) Evaluate $(1 + \omega - \omega^2)^8$

(xi) When the polynomial $x^3 + 2x^2 + kx + 4$ is divided by x – 2, the remainder is 14. Find value of k.

(xii) If α , β are the roots of $3x^2 – 2x + 4 = 0$, then find the value of $1/\alpha^2 + 1/\beta^2$

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Write only partial fraction form of $(x^2 + 1)/(x^3 + 1)$ without finding constants.

(ii) Resolve $(7x + 5)/((x + 3)(x + 4))$ into partial fraction.

(iii) Find the 13th term of the sequence x , 1 , 2 – x , 3 – 2x , ...

(iv) Show that reciprocals of the terms of the geometric sequence a_1 , a_1r^2 , a_1r^4 , ... form another geometric sequence.

(v) If $y = 1 + x/2 + x^2/4 + \ldots \ldots$ show that $x = 2((y – 1)/y)$

(vi) Find the nth term of H.P. 1/2 , 1/5 , 1/8 , ...

(vii) Write $(n + 1)(n)(n – 1)/(3 \cdot 2 \cdot 1)$ in the factorial form.

(viii) Find the value of n when ${}^nP_4 : {}^{n-1}P_3 = 9:1$

(ix) Find the number of diagonals of a 6-sided figure.

(x) Prove the formula $1 + 2 + 4 + \ldots \ldots + 2^{n-1} = 2^n – 1$ for n = 1 , 2

(xi) Using binomial theorem, expand $(a + 2b)^5$

(xii) Expand $(2 - 3x)^{-2}$ upto 4-terms.

Q4. (Marks 18) Attempt any NINE parts:

(i) Find θ when ℓ = 1.5cm ; r = 2.5cm

(ii) Verify sin60° cos30° – cos60° sin30° = sin30°

(iii) Prove that $2\cos^2\theta - 1 = 1 - 2\sin^2\theta$

(iv) If α, β and γ are the angles of triangle ABC then prove that sin(α + β) = sin(γ)

(v) Prove that $\cos(\alpha + 45°) = 1/\sqrt{2}\ (\cos\alpha - \sin\alpha)$

Note: (Continued P/2)

(vi) Express 2sin(3θ) cosθ as sum or difference.

(vii) Find the period of tan(x/7)

(viii) Find the value of sin 53°40'

(ix) Find area of triangle ABC, if a = 200 , b = 120 and γ = 150°

(x) Find the value of α if a = 7 , b = 7 , c = 9

(xi) Show that $\tan(\sin^{-1}x) = x/\sqrt{(1 - x^2)}$

(xii) Solve the equation sinx = 1/2

(xiii) Find the solutions of cot θ = $1/\sqrt{3}$; θ ∈ [0 , 2π]

Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5(a). (Marks 05) Show that $|x\ 1\ 1\ 1;\ 1\ x\ 1\ 1;\ 1\ 1\ x\ 1;\ 1\ 1\ 1\ x| = (x + 3)(x - 1)^3$

Q5(b). (Marks 05) Prove that $x^2/a^2 + (mx + c)^2/b^2 = 1$ will have equal roots if $c^2 = a^2m^2 + b^2$

Q6(a). (Marks 05) Resolve into partial fraction: 1/((x – 1)(2x – 1)(3x – 1))

Q6(b). (Marks 05) For what value of n, $(a^n + b^n)/(a^{n-1} + b^{n-1})$ is the positive geometric mean between a and b.

Q7(a). (Marks 05) Prove that: ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^nC_r$

Q7(b). (Marks 05) If $y = 1/3 + 1.3/2!(1/3)^2 + 1.3.5/3!(1/3)^3 + \ldots\ \ldots$ then prove that $y^2 + 2y - 2 = 0$

Q8(a). (Marks 05) If cosecθ = $(m^2 + 1)/2m$ and $m > 0 (0 < \theta < \pi/2)$ find the values of the remaining trigonometric ratios.

Q8(b). (Marks 05) Prove without using calculator $\sin 19° \cos 11° + \sin 71° \sin 11° = 1/2$

Q9(a). (Marks 05) Show that $r_3 = 4R \cos \alpha/2 \cdot \cos \beta/2 \cdot \sin \gamma/2$

Q9(b). (Marks 05) Prove that $\sin^{-1} 5/13 + \sin^{-1} 7/25 = \cos^{-1} 253/325$